

IAM ROLES AND POLICIES

Class no:3

1.IAM Policy:

An IAM policy is a JSON document that specifies permissions. Policies can be reused with different services in AWS. The same policy can be assigned to different people and teams. In AWS IAM policies are the backbone of managing permissions. These policies specify what actions are allowed or denied on AWS resources, helping you enforce security and control.

Types of policy:

- Identity based policy
 - Managed policy:
 - a. AWS Managed policy
 - b. Customer Managed policy
 - Inline policy (1:1 Relationship; STS [IAM role])
- Resource based policy

1.1 Identity based policy

A type of policy in AWS (Amazon Web Services) that specify permissions for IAM (Identity and Access Management) identities, such as users, groups, or roles. These policies determine what actions the identity can perform on specific AWS resources.

Eg: an identity-based policy can allow a user to launch EC2 instances or access S3 buckets.

Managed policy:

- **AWS Managed Policies:** These are the policies that are provided by default in AWS. These consist of the commonly used services and the read and write permissions associated with them.
- **Customer-Managed Policies:** In case a user needs a policy that should meet their specific needs and use cases, then they can build their custom policies.

Inline Policy

Inline policies are directly embedded into a single IAM user, group, or role. They offer a one-to-one relationship, meaning that the policy is tied specifically to that entity. While inline policies provide granular control, they are not reusable across different users or roles, making them less flexible than managed policies.

1.2 Resource based policy

Unlike the first two types, which are attached to users, roles, or groups, resource-based policies are attached directly to AWS resources, such as S3 buckets, Lambda functions, or SNS topics. These policies define which principals (users, roles, or services) are allowed to access the resource and what actions they can perform. Resource-based policies are useful for controlling cross-account access to resources.

Hands on: Creating a user and adding the different types of policies:

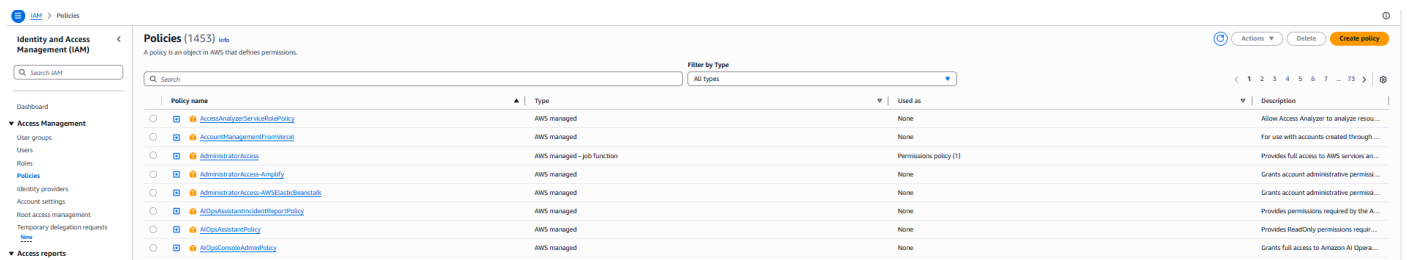
(i)To add amazon managed policies:

Step 1: Open IAM, click user, then create use, Create an IAM user named Nithya.

Step 2: Then click on Nithya, choose add permissions, then choose Attach policies directly. We can see the list of amazon manged policies which are inbuilt in nature. Choose AmazonS3fullaccess.

(iii) To add Customer managed policies:

We have to choose policies in IAM, then choose create policy there. These policies can be accessed in the attach direct policy. Instead of choosing AWS managed policy, scroll down and choose our customer based policy there.



2.IAM ROLES:

AWS IAM Roles allow users, applications and services to securely access resources without requiring permanent credentials. Unlike IAM users, which are assigned to specific individuals, roles can be temporarily used by AWS services like EC2 and Lambda or even by external entities. This approach helps businesses manage permissions more efficiently while ensuring security best practices, such as granting access only when necessary. We can perform,

(i) **Role based:** service to service connection.

(ii) **Account based:** temporary access activities.

Hands on: Creating a role and adding policies:(account based)

Step 1: Go to IAM, click roles, then create new role.

Choose: **AWS account** and **enable** the respective account id. Then click on next.

Select trusted entity [Info](#)

Trusted entity type

☐ AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☒ AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

An AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☒ This account (314383175192)

☐ Another AWS account

Options

☐ Require external ID (Best practice when a third party will assume this role)

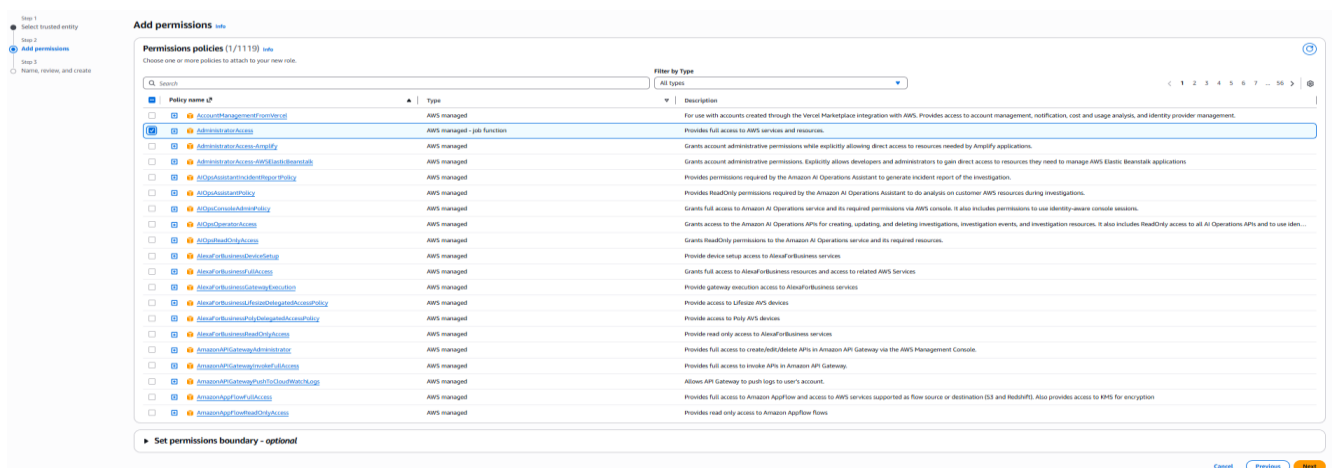
☐ Require MFA

Requires that the assuming entity use multi-factor authentication.

[Cancel](#)

[Next](#)

Step 2: Choose the policies we want to add and click on next (Here I am providing admin access policy to this role).



Step 3: Name the new role: devopsrole1, click create role.

The screenshot shows the 'Name, review, and create' step in the AWS IAM console. The 'Role details' section has 'devopsrole1' entered for the role name and a description. Below, the 'Trust policy' section shows a JSON policy for assuming the role. The 'Permissions policy summary' section shows the 'AdministratorAccess' policy is attached. The 'Add tags' section is empty. At the bottom right are 'Cancel', 'Previous', and 'Create role' buttons.

In order to access the new role devopsrole1 we need to add an inline policy known as **STS (Security Token Service) to our user: Nithya**. It is a web service provided by Amazon Web Services that allows you to request temporary, limited-privilege credentials for AWS Identity and Access Management (IAM) users or for federated users you authenticate.

Step 4: Go to user, click our user Nithya and click Add permissions, choose: create inline policy.

The screenshot shows the 'Nithya' user page in the AWS IAM console. The 'Summary' section displays the user's ARN, creation date, and console access status. The 'Permissions' tab is selected, showing a table with one policy: 'IAMUserChangePassword' (AWS managed, attached directly). Buttons for 'Add permissions', 'Add permissions', and 'Create inline policy' are visible on the right. At the bottom are 'Permissions boundary' and 'Generate policy based on CloudTrail events' options.

Step 5: Choose STS as the resources we want to add, the actions we want to allow, Resources as all, then click next. (Without the STS, the user cannot access the role we have created.)

The screenshot shows the 'Add permissions' dialog in the AWS IAM console. The 'Actions allowed' section has 'All STS actions (sts:*)' selected. The 'Resources' section has 'All' selected. The 'Effect' is set to 'Allow'. At the bottom are 'Add more permissions', 'Cancel', and 'Next' buttons.

Step 6: Provide the policy a name: devopsrole1Policy, then click create policy.

Review and create [Info](#)
Review the permissions, specify details, and tags.

Policy details

Policy name
Enter a meaningful name to identify this policy.

Maximum 128 characters. Use alphanumeric and '+', '@', '-' characters.

Permissions defined in this policy [Info](#) [Edit](#)
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Allow (1 of 464 services) [Show remaining 463 services](#)

Service	Access level	Resource	Request condition
STS	Full access	All resources	None

[Cancel](#) [Previous](#) [Create policy](#)

Step 7: Now to access the role using our IAM user, Open the **IAM user account: Nithya console**, on the top right click on the account id, we can see the **switch role** option click on it.

The screenshot shows the AWS IAM console interface. On the right side, a dropdown menu is open, displaying account information and a 'Switch role' button. The account information includes: Account ID (3143-8317-5192), Account name (Access denied), Account color (Access denied), and IAM user (Nithya). The 'Switch role' button is highlighted. The background shows the 'Console Home' with sections for 'Recently visited', 'Applications (0)', 'Welcome to AWS', 'AWS Health', and 'Cost and usage'.

Step 8: It will ask for the Account id and user name of the role. Enter and click switch role.

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID
The 12-digit account number or the alias of the account in which the role exists.

IAM role name
The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`.

Display name - optional
This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

Display color - optional
The selected color displays in the console navigation when this role is active
☐ None

[Cancel](#) [Switch Role](#)

Step 9: We are now switched to the devopsrole1 role. The confirmation can be done by clicking the Account id on the top, which shows the current active role.

As we have provided the admin access policy in the role, this role allows us to perform all actions.

The screenshot displays the AWS Management Console interface. The top navigation bar shows the AWS logo, a search bar, and the current region (Europe (Stockholm)). The main content area is divided into several sections:

- Console Home**: A central section with a "Recently visited" card showing "No recently visited services" and a list of common services (EC2, S3, CloudWatch, IAM). Below this are cards for "Welcome to AWS", "AWS Health", and "Cost and usage".
- Applications (0)**: A section for managing applications, currently showing no results.

A dropdown menu is open on the right side of the console, displaying the following information:

- Currently active as**: devopsrole1
- Account ID**: 3143-8317-5192
- Account name**: Nithyavel
- Account color**: Unset
- Signed in as**: Nithya
- Account ID**: 3143-8317-5192
- Role history**: devopsrole1 @ 314383175192

The dropdown menu also includes a "Switch back" button and a "Role history" section.